CoReLab
COMPUTATIONAL RESEARCH LABORATORY
University of Stavanger

MASTER THESIS

Investigation of Sodium Insertion into Black Phosphorus Electrode: A Molecular Dynamics study

Rechargeable batteries play a key role in the transition to renewable energy, from electric vehicles to grid-scale energy storage. Sodium-ion batteries are emerging as a promising alternative to lithium-ion batteries, as sodium is more abundant, potentially reducing costs and supply chain risks. However, further research is needed to improve their competitiveness. Novel electrode materials such as black phosphorus offer the potential to significantly increase the energy density of sodium-ion batteries. However, the mechanisms governing sodium insertion are not yet fully understood. Understanding these processes at the atomic scale is essential for developing safer, higher-performing electrodes.

In this project, you will explore these mechanisms using computational techniques such as Molecular Dynamics and Rare Event Sampling. You will select suitable simulation tools, construct molecular systems, and simulate the motion and interactions of individual atoms and molecules. How cool is that! Finally, you will apply statistical physics to analyze your results and gain insights into the underlying physical processes.

> **Bonus:** If the results are good, they can become part of a scientific publication, with you as a co-author!

What's in store for you

- **Computational Skills:** Gather profound experience with advanced molecular modeling techniques, independently perform computational simulations and enhance your computational skills.
- **Analytical Work:** Evaluate your data with modern analysis procedures and create valuable insights into molecular phenomena.
- **Purpose:** Be part of a research group, see how we work and make real scientific contributions to battery research.

Supervisor
Jonas Jaeger
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Thesis type
Master thesis, 30/60 ECTS

Start
Spring 2027 or by agreement

More topics

corelab-uis.github.io/Website

Your profile

- **Background:** Computational Engineering, Physics, Math, Chemistry, Mechanical Engineering or comparable.
- **Skills:** Good skills in python coding and solid understanding in thermodynamics required. Prior experiences with Linux and molecular modeling techniques are advantageous but not a necessity.
- **Motivation:** You show initiative, hold high quality standards and work responsibly.

Interested?

Send a short email with your CV and transcript to the supervisor. We are happy to discuss the topic and adapt it to your interests.